
Integrated Management Solutions For Durga Traders' Challenges

Kratika Varshney [23f2001844]



**BUSINESS DATA MANAGEMENT
PROJECT PRESENTATION**

Overview

Durga Traders is a B2B coal wholesaler based in Aligarh, Uttar Pradesh, supplying products such as Lam Coke, Met Coke, Charcoal, and Pig Iron to industrial depots across multiple cities. The firm sources coal from regions like West Bengal, Haldia, and Orissa and primarily operates on credit-based transactions.



PROBLEM STATEMENT & OBJECTIVES

Problem Statement

- Fluctuating purchase prices lead to unstable profit margins.
- Credit-heavy sales delay cash inflows and strain liquidity.
- Coal quality deteriorates during monsoon due to moisture absorption.

Objectives

- Analyze profit margin variability across products.
- Evaluate credit structure and liquidity position.
- Study impact of coal quality on inventory and margins.
- Propose practical, data-driven business solutions.

Nature of Data Collected

Data Period : April 2022- March 2023

Data Sources :

- Purchase Ledger
- Sales Ledger
- Debtors Sheet
- Creditors Sheet
- Balance Sheet

Tools Used : Google Sheets (Pivot tables, ratio analysis, charts).

Data Cleaning & Descriptive Stats

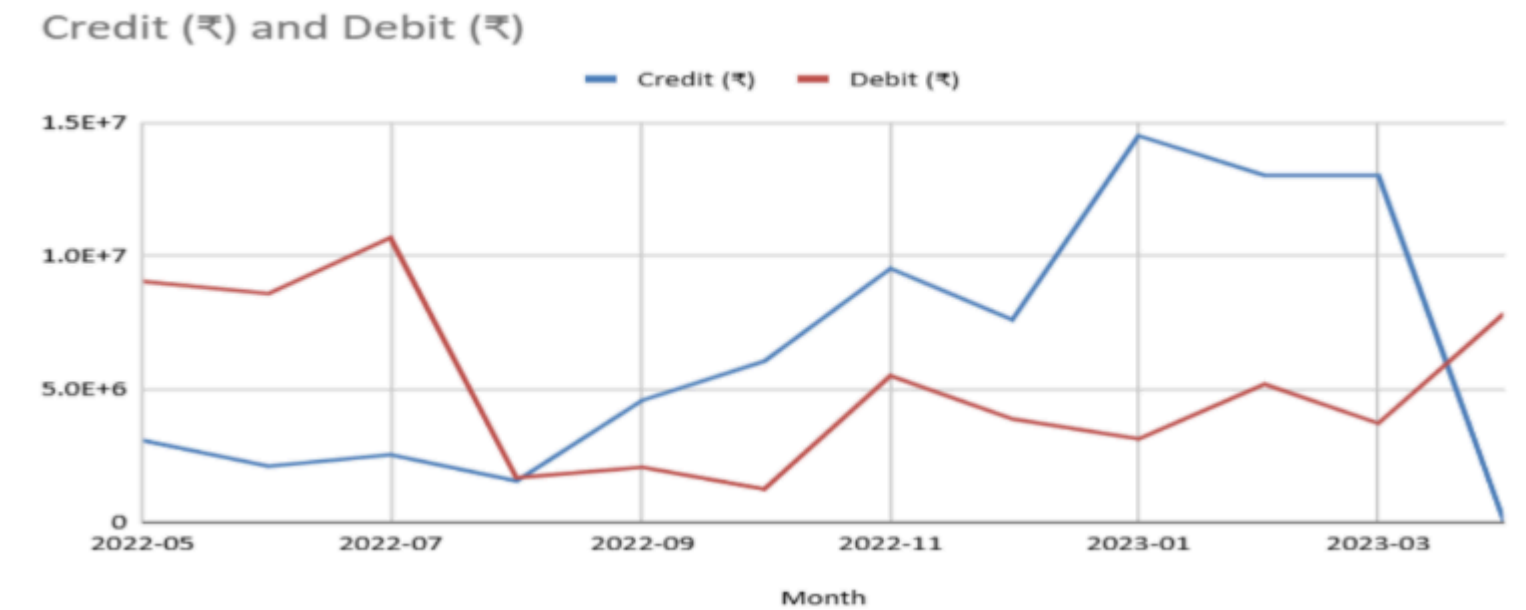
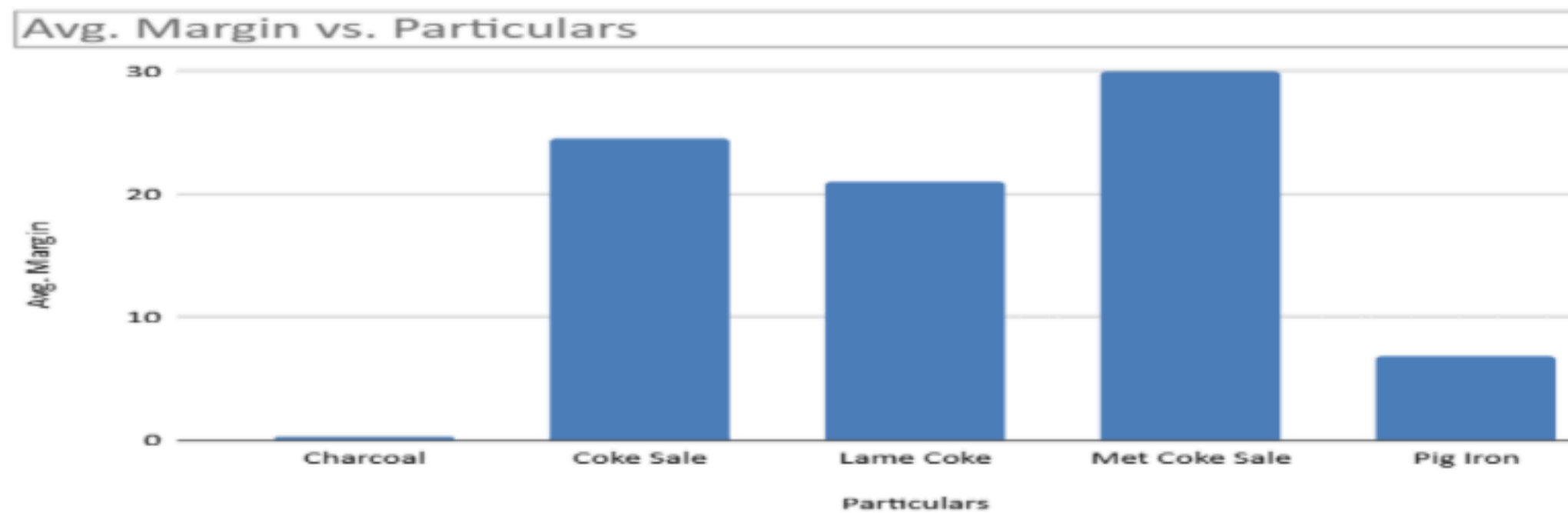
Data Cleaning & Preparation

- Standardized product names.
- Converted text to numeric format.
- Created Profit Margin %, Monthly aggregates.
- Cross-verified totals with balance sheet.

Descriptive Statistics

- Mean Monthly Inflow $\approx$ ₹70.5 lakh
- Mean Monthly Outflow $\approx$ ₹56.8 lakh
- Average Monthly Surplus $\approx$ ₹13.7 lakh
- High variability observed across months due to seasonal demand and bulk purchases.

ANALYSIS (PROFIT & SALES)



- Met Coke & Coke Sale show highest margins.
- Charcoal & Pig Iron generate lowest returns.
- Margins decline during monsoon due to quality deterioration.

ANALYSIS (CREDIT & CASH FLOW)

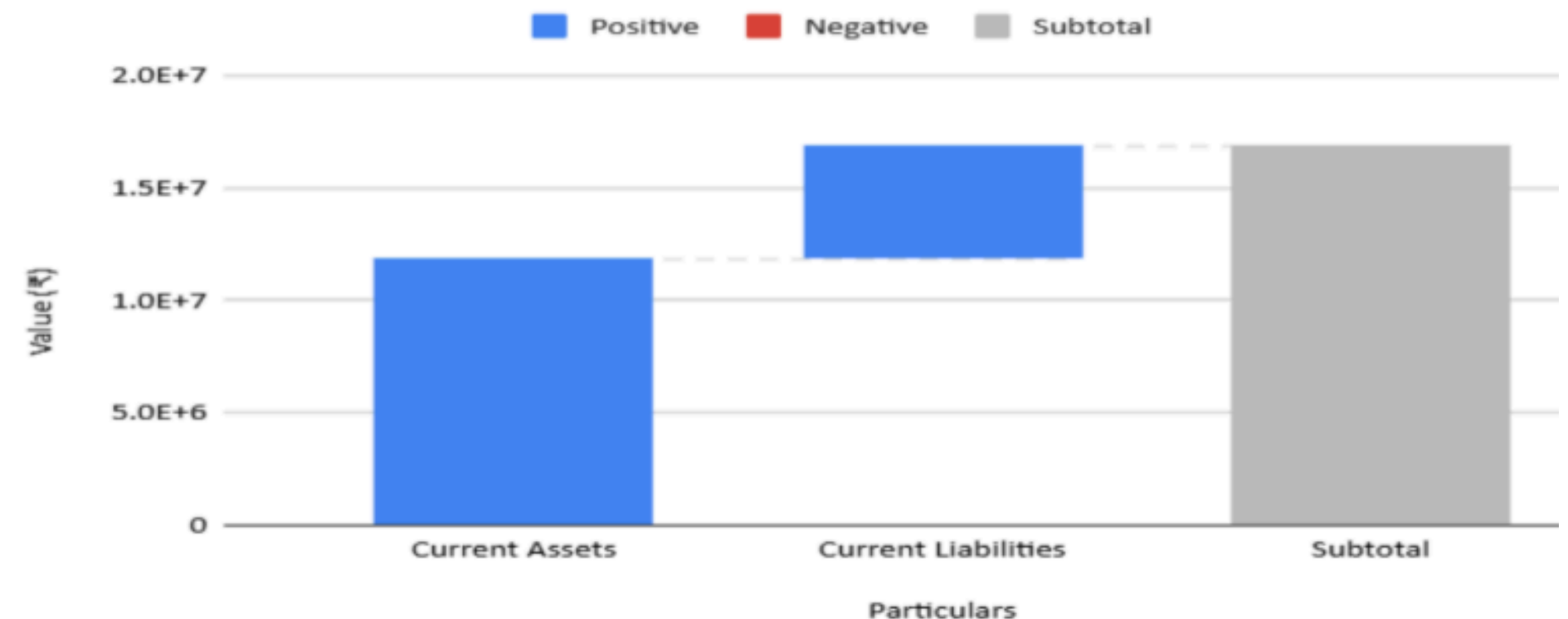
Comparison of Debtors and Creditors



Creditors are significantly higher than debtors, showing heavy reliance on supplier credit and delayed cash realization.

Assets exceed liabilities, indicating strong liquidity on paper, but cash flow remains strained due to credit-based sales.

Value (₹) vs. Particulars



Insights

- Profit margin instability driven by supplier price fluctuations.
- Monsoon moisture reduces coal quality and selling price.
- Credit-heavy sales delay cash realization.
- High dependency on supplier credit increases financial risk.
- Liquidity strong on paper, but operational cash flow remains strained.

Recommendations

Stabilize Profit Margins

- Implement 6–12 month supplier rate contracts to reduce price volatility.
- Adopt cost-linked dynamic pricing to maintain 20–30% target margins.
- Monitor product-wise margin monthly through dashboard tracking.

Strengthen Cash Flow & Credit Control

- Introduce customer credit scoring to regulate credit limits.
- Offer 2% early-payment incentive to accelerate receivables.
- Maintain receivable days lower than payable days to avoid liquidity mismatch.

Reduce Quality & Seasonal Risk

- Invest in covered storage to prevent moisture damage during monsoon.
- Reduce monsoon procurement and prioritize dry-stock clearance.
- Track supplier quality performance through batch-wise monitoring.



THANK YOU!